

ST LOUIS DOWNTOWN, IL, USA

WMO: 725314

Lat: 38.571N

Lon: 90.157W

Elev: 413

StdP: 14.48

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 03960

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	8.2	12.6	-2.6	4.9	10.8	1.2	5.9	14.8	26.1	28.0	23.7	29.3	8.4	310	0.458

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.9	95.1	76.6	92.6	76.2	90.5	75.3	79.8	90.5	78.3	88.9	77.1	87.2	7.4	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
76.9	142.1	85.6	75.1	133.9	84.0	73.5	126.6	82.3	43.5	91.1	42.0	89.4	40.7	87.2	85.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
20.8	18.6	16.6	0.7	99.5	5.5	3.0	-3.3	101.6	-6.5	103.4	-9.6	105.0	-13.5	107.2		
			0.0	82.5	5.2	1.7	-3.8	83.7	-6.8	84.7	-9.7	85.7	-13.5	86.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	56.7	32.6	36.1	46.2	57.2	67.0	75.0	78.7	76.9	69.6	58.1	46.0	36.0
	DBStd	18.49	11.30	11.07	10.90	9.03	8.01	5.98	5.50	5.28	7.25	9.22	9.67	10.32
	HDD50	1857	548	407	207	34	1	0	0	0	0	27	187	446
	HDD65	4512	1004	810	590	263	77	6	0	1	37	254	572	898
	CDD50	4313	10	17	87	249	527	751	889	835	589	278	68	13
	CDD65	1493	0	0	6	28	138	307	424	371	176	41	2	0
	CDH74	15128	0	3	52	294	1205	3114	4656	3656	1729	394	25	0
	CDH80	6238	0	0	4	60	364	1297	2164	1574	670	103	2	0
Wind	WSAvg	6.9	8.4	8.7	8.7	8.6	6.8	5.8	5.1	4.8	5.0	6.4	7.4	7.8
Precipitation	PrecAvg	39.1	2.1	2.1	3.5	4.0	4.3	4.3	3.7	3.0	3.1	2.9	3.3	2.8
	PrecMax	61.3	9.0	5.2	8.4	10.4	12.9	13.1	10.7	6.7	9.8	12.4	10.0	11.7
	PrecMin	23.5	0.1	0.2	1.1	1.0	0.8	0.4	0.1	0.1	0.0	0.2	0.4	0.7
	PrecStd	8.0	1.6	1.2	1.5	2.3	2.4	2.7	2.2	1.6	2.2	1.9	2.0	2.0
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.5	71.5	79.4	85.7	90.7	95.5	100.2	97.9	94.1	88.1	77.8	67.8
		MCWB	58.3	60.2	63.3	67.1	72.3	74.9	76.6	77.1	73.9	71.6	64.1	60.4
	2%	DB	60.7	65.5	74.1	81.3	87.3	92.6	96.2	94.0	90.8	82.4	71.7	62.0
		MCWB	53.7	56.0	61.9	65.2	71.1	75.2	77.8	77.1	73.9	67.6	59.8	55.8
	5%	DB	55.0	59.6	69.4	77.2	84.1	90.4	92.9	91.0	87.8	78.8	66.3	56.4
		MCWB	49.2	52.0	58.4	63.1	69.8	74.5	77.2	76.0	72.9	65.4	57.7	50.6
	10%	DB	49.5	54.2	64.3	73.1	81.2	87.9	90.5	88.3	83.6	73.6	62.7	51.7
		MCWB	43.9	47.4	55.8	61.2	68.6	73.3	76.4	74.6	70.2	63.3	55.2	46.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.2	61.9	66.4	70.2	76.2	79.3	82.2	81.7	77.8	73.1	65.8	62.4
		MCDB	65.4	69.5	76.5	81.2	85.6	90.2	91.7	90.6	88.9	84.2	74.2	66.7
	2%	WB	55.2	58.0	63.0	67.5	73.9	77.5	80.2	79.4	76.0	70.7	62.6	57.4
		MCDB	58.9	64.0	71.1	77.2	83.5	88.9	90.9	89.9	86.8	79.2	68.1	61.5
	5%	WB	49.7	52.8	60.2	65.2	71.9	76.2	78.8	77.8	74.3	68.2	59.4	51.7
		MCDB	55.1	58.2	67.7	73.9	80.5	87.0	89.7	88.1	83.5	75.1	65.4	54.8
	10%	WB	43.7	46.9	56.9	62.9	70.3	74.9	77.6	76.3	72.6	65.4	55.5	46.6
		MCDB	49.2	53.4	63.7	71.1	78.4	84.7	87.7	85.5	80.6	72.6	61.9	50.3
Mean Daily Temperature Range	5% DB	MDBR	16.5	17.6	19.7	21.2	19.9	20.1	19.9	20.7	23.1	22.0	19.5	15.8
		MCDBR	25.6	27.8	25.9	26.6	22.8	22.9	22.2	23.3	26.1	26.6	26.4	22.9
		MCWBR	18.9	19.6	15.7	13.9	10.5	8.9	8.4	9.2	10.9	13.3	17.2	17.1
	5% WB	MCDBR	23.3	24.8	22.7	22.1	19.2	19.9	19.8	20.3	21.3	21.5	22.2	19.6
		MCWBR	18.9	19.4	14.9	13.5	10.3	8.9	8.7	9.3	9.7	12.1	16.7	16.9
Clear-Sky Solar Irradiance	taub	0.307	0.317	0.350	0.403	0.439	0.449	0.476	0.467	0.411	0.352	0.312	0.303	
	taud	2.472	2.433	2.402	2.249	2.213	2.216	2.143	2.163	2.300	2.468	2.522	2.496	
	Ebn at Noon	277	289	288	277	269	265	256	255	264	272	273	270	
	Edh at Noon	25	30	34	43	45	45	48	46	38	29	24	23	
All-Sky Solar Radiation	RadAvg	650	876	1187	1571	1762	2016	1970	1769	1505	1087	736	546	
	RadStd	72	90	116	90	156	146	136	101	96	118	77	57	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+254
(47)	Regional (33 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+144

Nomenclature: See separate page